

RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR

COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN
HIGHER NATIONAL DIPLOMA EXAM (HND)

REPUBLIC OF CAMEROON
Peace-Work-Fatherland

MINISTRY OF HIGHER EDUCATION

NATIONAL COMMISSION FOR THE ORGANISATION
OF HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM

National Exam of Higher National Diploma– 2023 Session

Spécialty/option : SWE. CSN. DBM

Paper : Digital Electronics

Duration : 4hours

Credit : 7/74

Instructions: Answer all questions. Calculators of any type are not authorized!

SECTION A: NUMBER SYSTEM AND CODES (20marks).

MCQs, for each question, four answers are proposed. Write the letter for the correct answer against the question number. Each question carries 1mark.

- 1) The value of radix in binary number system is _____
 - a) 2
 - b) 8
 - c) 10
 - d) 1
- 2) Which of the following is not a binary number?
 - a) 1111
 - b) 101
 - c) 11E
 - d) 000
- 3) Positive integers must be represented by
 - a) signed numbers
 - b) unsigned number
 - c) Both A and B
 - d) None of above
- 4) An unweight code in which only one bit changes from one code number to the next is:
 - a) BCD
 - b) Excess-3
 - c) Gray
 - d) ASCII

- 5) In the decimal numbering system, what is the MSD?
- a) The middle digit of a stream of numbers
 - b) The rightmost digit to the decimal point of the number ✓
 - c) The last digit on the right ✓
 - d) The digit with the smallest weight
- 6) The following BCD number: 0010 1000.0111 0100 corresponds to the decimal number:
- a) 28.72 ✓
 - b) 29.52
 - c) 28.70 ✓
 - d) 28.82 ✓
- 7) How many binary digits are required to count to 100_{10} ?
- a) 100
 - b) 3
 - c) 7
 - d) 2
- 8) Which of the following logical operations is represented by the + sign in Boolean algebra?
- a) Inversion
 - b) AND
 - c) OR
 - d) complementation
- 9) The binary equivalent of gray code 1111 is :
- a) $(1101)_2$
 - b) $(1011)_2$
 - c) $(1110)_2$
 - d) $(1010)_2$
10. The BCD number for decimal 347 is _____.
- a) 1100 1011 1000
 - b) 0011 0100 0111
 - c) 0011 0100 0001
 - d) 1100 1011 0110

11. Convert binary to octal: $(110110001010)_2 = ?$

- a) $(5512)_8$
- b) $(6612)_8$
- c) $(4532)_8$
- d) $(6745)_8$

12. What is the addition of the binary numbers 11011011010 and 010100101?

- a) 0111001000
- b) 1100110110
- c) 11101111111
- d) 10011010011

13. Binary subtraction of $100101 - 011110$ is

- a) 000111
- b) 111000
- c) 010101
- d) 101010

14. Perform multiplication of the binary numbers: 01001×01011

- a) 001100011
- b) 110011100
- c) 010100110
- d) 101010111

15. Divide the binary numbers: $1010001 \div 1000$

- a) 1010.001
- b) 1000.001
- c) 1100.001
- d) 0011.001

16. Which numbering system uses numbers and letters as symbol?

- a) Decimal
- b) binary
- c) Octal
- d) Hexadecimal

17. The octal number $(651.124)_8$ is equivalent to _____

- a) $(1A9.2A)_{16}$
- b) $(1B0.10)_{16}$
- c) $(1A8.A3)_{16}$
- d) $(1B0.B0)_{16}$

18. The sign of a negative binary number is given by by its _____

- a) MSB
- b) LSB
- c) Byte
- d) Nibble

19. Which of the following is a correction detection code?

- a) Gray code
- b) Hamming code
- c) UNICODE
- d) Parity code

20. The format used to present the logic output for the various combinations of a logic expression is called:

- a) Boolean constant
- b) Boolean variable
- c) Truth table
- d) Logic function

SECTION B: COMPUTER FUNDAMENTAL (80marks)

I) Hardware (20 Marks)

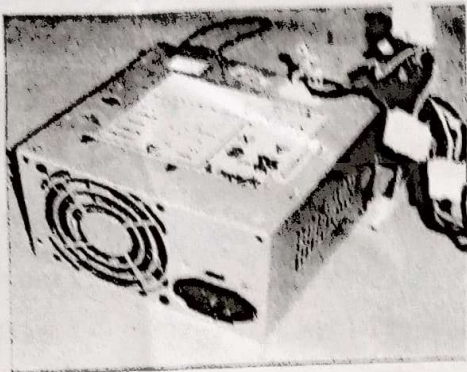
1) Define the following acronyms:

- a) RAM
- b) SDRAM
- c) EEPROM
- d) ROM
- e) DRAM

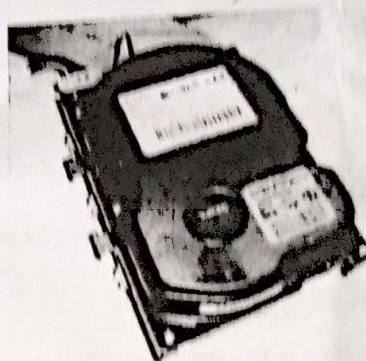
(5×1 = marks)

2) Name the following devices and give their roles

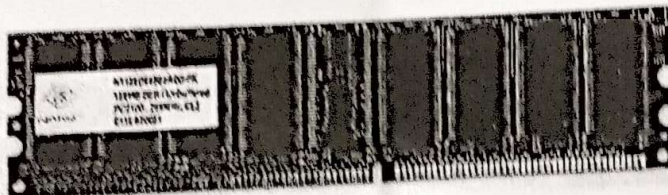
a)



b)



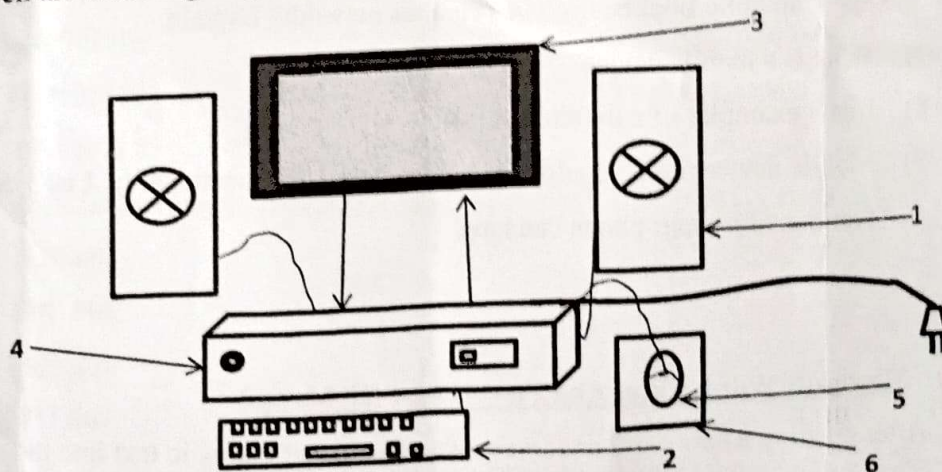
c)



(3 × 2 = 6 marks)

3) What is the main difference between a mainframe computer and a supercomputer?
(2marks)

4) Given the following diagram below:



(3marks)

a) 4. a) Identify the various components from 1 to 6.

(4marks)

b) 4. b) Identify two input devices and two output devices.

II) Network And Mobile Devices (40 Marks)

- 1) Give a brief description of the each of the following network devices
 - a) Router
 - b) Switch
 - c) Access point
 - d) Firewall
 - e) Repeater

(5 x 2 marks)
- 2) What is the Internet?

(2 marks)
- 3) What is the difference between Internet and internet?

(2 marks)
- 4) A company might decide to invest in an intranet. What are the benefits of this type of network?

(2 marks)
- 5) How is Internet different from intranet?

(2 marks)
- 6) Network can be classified using various criteria.
 - a) Classify networks based on geographical distances.

(5 marks)
 - b) Classify networks based on the media of transmission.

(3 marks)
 - c) What network type is suitable for a single school campus?

(1 mark)
 - d) What transmission media is preferable for the network in (c) above.

(1 marks)
 - e) Can optic fiber be used in a campus network? Explain.

(2 marks)
- 7) What is a mobile device?

(2 marks)
- 8) List 3 examples of a mobile device.

(3 marks)
- 9) Mobile devices come equipped with hardware for functionality. List 5 common features of a smart phone can have

(5 marks)

III) Microsoft Word, Excel And PowerPoint (20 Marks)

- 1) After creating a new word document, a user can start typing in text into the new work space
 - a) What is the default tab of a new document?

(1 mark)
 - b) What file extension does a word document save in?

(1 mark)
 - c) Which tab is used to add illustrations to a word document

(1 mark)
 - d) What is the role of the format painter in a Word document?

(2 marks)

2) Consider the following figure below.

2		CA Results	Exam Results	Total
3	1	15	48	
4	2	16	39	
5	3	8	36	
6	4	22	57	
7	5	17	43	
8	6	26	60	
9				
10				

- Which Microsoft packages can be used to display the above table? (2 marks)
- Which of the packages listed in (a) above contains dynamic squares? (1 mark)
- What is the name given to each square? (1 mark)
- What is the default data type of each dynamic box? (1 mark)
- Suppose a teacher wants to obtain the totals of the students' scores, what steps are taken to quickly do this? (5 marks)

3) Graphic designers might use Microsoft office tools for productivity

- Which Microsoft tools can be used to design a flier (3 marks)
- What file format does a PowerPoint file save in? (2 mark)